

MATTHEW ESPOSITO

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Education

William & Mary

Expected May 2026

BS in Computer Science (Cybersecurity path) & BA in Philosophy (Ethics for Data & Computing path) Williamsburg, VA

- Awards: 2x W&M Cypher Hackathon Award, [Cyber NCL](#) team placed **2nd** (Spring '25), **6th** (Fall '24) nationwide

Experience

Memfault

May 2025 – August 2025

Software Engineering Intern

Boston, MA

- Added features to high volume backend processing pipeline in Python, retaining performance and correctness
- Extended Linux SDK to support [custom traces](#), allowing for more flexible error reporting, in Rust
- Created [Python library](#) to intercept unhandled exceptions and report them to Memfault platform, enabling plug-and-play error collection and observability for Python customers
- Implemented full unit and integration tests, extended BitBake recipes and QEMU test images to verify custom trace implementation
- Feature implementations in high-volume data ingestion pipeline

Memfault

May 2024 – August 2024

Software Engineering Intern

Boston, MA

- Designed and implemented new feature to parse and report stack traces in Rust and Python
- Trained a machine learning model to categorize embedded hardware's stack traces and core dumps into common fault categories, achieving >95% accuracy and helping customers more rapidly fix problems.
- Identified and resolved CI/CD bottlenecks, reducing run times and cutting costs.
- Created tool to integrate CRM tools with Algolia APIs to measure engagement in high-potential target accounts.

Geospatial Evaluation and Observation Lab

January 2023 – Present

Software Developer, Team Lead

Williamsburg, VA

- Productionized [ICEBREAKER](#) project, a political wargaming evaluation tool backed by RAG-enabled LLM
- Built SCOPE, a news info extraction/summarization tool with ML and LLMs.
- Led team and developed Kubernetes deployments and CI/CD pipelines.
- Implemented data ingestion pipeline for SCOPE, enabling real-time data processing and analysis via LLMs.

Pex (Remote)

December 2023 – April 2024

Software and Hardware Developer

- Developed embedded IoT prototypes from inception to testing in 12 weeks
- Architected solutions with C++, embedded systems, sound protocols, and cellular modem development
- Wrote C++ library for integrating with SIM7600 cellular modem for sub-30ms latency HTTPS/MQTT communication

Trilium.cc

December 2021 – Present

Founder

- Starting a Software-as-a-service business and Linux server management.
- Serving over 150 customers and maintaining 99.9% uptime while learning backend development.
- Automated system tasks including backups, updates, and new server deployments with Ansible.

Projects

[ALPR Analysis](#) | *Rust, GIS, OSINT*

Dec 2025 – Present (maintenance)

- Engineered a high-throughput data ingestion pipeline in Rust to process planetary-scale OpenStreetMap datasets (60GB+) for surveillance infrastructure density + traffic flow analysis.
- Optimized performance using parallel processing and memory-efficient streaming parsers, enabling rapid analysis of planet-scale geospatial data.
- Designed a heuristic inference engine to classify map nodes with high precision, leveraging vendor fingerprinting and contextual metadata.
- Utilized optimized graph contraction hierarchies to speed up fast-path lookup, from every home in the USA to its nearby amenities
- Built and deployed a Vue.js visualization dashboard to map surveillance density, transforming raw OSINT data into actionable insights.

- Maintainer of open-source Reddit frontend (2.5k+ stars, 1.7M+ container pulls/mo, 100k+ hits/day on large instances)
- Enabled public access for censorship-heavy regions, bulk academic research.
- Reverse-engineered Reddit mobile API for unthrottled data access for public benefit.
- Implemented peer-to-peer coordination protocol to handle instance blockages live, based on HyParView/PlumTree epidemic broadcast trees, as part of a project for my Cloud Computing class.
- Performant backend development for very high-traffic Redlib instances, with proper caching/compression/lazy loading, as well as best practices for lightweight webpages
- Code review, OSS leadership, mentoring contributors

- Full-stack web & mobile app with hardware prototype (C++/LVGL).
- Rust-based backend handling DNS-over-HTTPS requests according to user-set schedule, 10k+ req/s.
- Built in 24-hour solo hackathon; won Best Hardware Hack.
- Wrote [blog post](#) about experience.

- Campus occupancy prediction via BLE beacon counting IoT devices and serverless stack.
- 90+% accuracy in long-term trend forecasting using Prophet + time series ML.
- [Writeup](#) featured on [Hackaday](#) and cited in [MIT Engineering courses](#).

Research

Vulnerability Testing Report

Sep 2024 – Dec 2024*Volunteer Cybersecurity Researcher**Williamsburg, VA*

- Performed a volunteer full security assessment of a local business's public mobile app; authored 22-page CVSS report.
- Discovered six serious security vulnerabilities, from advanced side channel methods to basic authentication failures
- Recommended remediations led to complete patching and mitigation.
- Disclosure coordinated; redacted report available on request.

Relevant Coursework

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|-------------------------|-------------------------|-----------------------------|----------------------------|
| • Algorithms | • Finite Automata | • Cloud Computing | • Ethics in Data Science |
| • Computer Organization | • Operating Systems | • Computer/Network Security | • Philosophy of Technology |
| • Software Development | • Applied Cybersecurity | | • Linear Algebra |

Technical Skills

Technical Areas: Backend Development, IoT, Embedded Systems, Systems Programming, DevOps, Docker, Git, Nginx, Prometheus, Grafana

Languages: Rust, C, C++, C#, Python, Bash, JavaScript

Interests: Open Source, Linux, Rust